

Phase Diagram of Noisy MNIST Classification: A Finite-Size Scaling Analysis of Corruption, Batch Size, and Dataset Size

1 Introduction

A fundamental question in supervised learning is: *under what conditions can a neural network extract useful signal from heavily corrupted training data?* This report documents a systematic computational study that maps out the **phase diagram** of a single-hidden-layer MLP trained on MNIST digits whose pixels have been subjected to salt-and-pepper noise. The analysis draws on concepts from statistical physics—phase transitions, critical exponents, and finite-size scaling—to characterize the sharp boundary between a “learning” phase (where the network achieves non-trivial test accuracy on clean data) and a “non-learning” phase (where performance remains at chance level, $\sim 10\%$).

The experiments sweep over three control parameters:

1. **Corruption probability** $p \in [0.90, 0.995]$: the fraction of input pixels replaced by uniform random values.
2. **Batch size** $B \in \{16, 32, 64, \dots, 4096, 60000\}$: spanning minibatch SGD through full-batch gradient descent.
3. **Training set size** $N \in \{5000, 10000, 20000, 40000, 60000\}$: used to probe finite-size scaling behavior.

For each (p, B, N) triplet the experiment is repeated over 5 random seeds, and the mean clean-test accuracy is recorded as the order parameter.

2 Experimental Setup

2.1 Model Architecture

The classifier is a single-hidden-layer MLP:

$$f(x) = W_2 \sigma(W_1 \text{vec}(x) + b_1) + b_2,$$

where $x \in \mathbb{R}^{28 \times 28}$ is the input image, $W_1 \in \mathbb{R}^{h \times 784}$, $W_2 \in \mathbb{R}^{10 \times h}$, σ is the ReLU activation, and the hidden width is $h = 512$.

2.2 Noise Model

Each training image undergoes **salt-and-pepper corruption**: every pixel is independently replaced with a sample from $\text{Uniform}(0, 1)$ with probability p , and left unchanged with probability $1 - p$. At $p = 0.90$, 90% of pixels are noise; at $p = 0.995$, only 0.5% of the original signal survives. Crucially, the test set is always evaluated *clean* (no corruption), so test accuracy measures whether the network has extracted genuine digit structure from the noisy training signal.

2.3 Training Protocol

All runs use SGD with learning rate $\eta = 0.01$ and no momentum or weight decay. The total computation budget is fixed at $N \times 50$ sample presentations (i.e., 50 “reference epochs”), regardless of batch size. This means that a run with $B = 64$ takes many more gradient steps than one with $B = 60000$ (full-batch), ensuring a fair comparison of the *information content per sample* rather than per step.

2.4 Parameter Grid

Parameter	Values
Corruption p	30 values linearly spaced in $[0.90, 0.995]$
Batch size B	16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 60000
Dataset size N	5000, 10000, 20000, 40000, 60000
Seeds	0, 1, 2, 3, 4

Table 1: Experiment parameter grid. Total configurations: $30 \times 10 \times 5 \times 5 = 7,500$ individual runs.

3 Results and Observations

3.1 Phase Diagram at Full Dataset Size

Figure 1 shows the primary result: a heatmap of mean clean-test accuracy in the (p, B) plane at $N = 60000$.

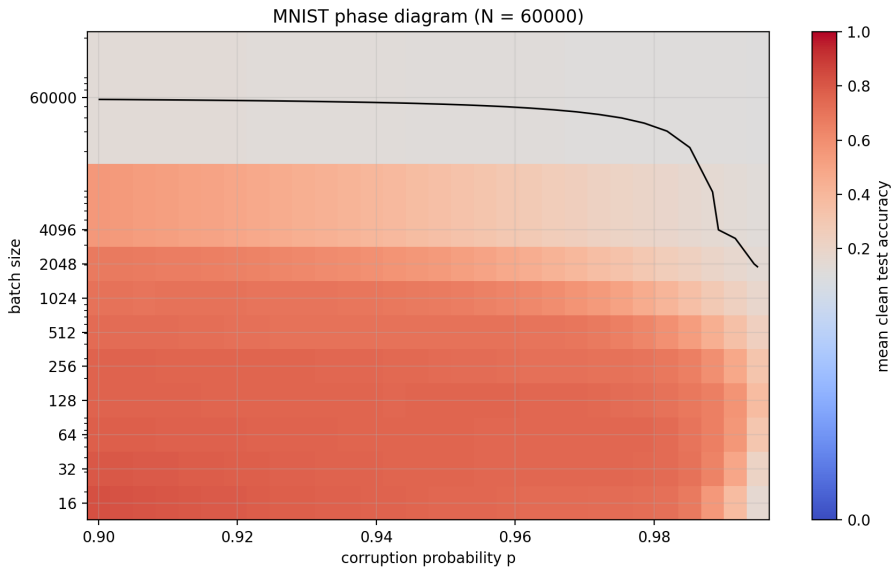


Figure 1: Phase diagram at $N = 60000$. Color encodes mean clean-test accuracy (blue = 1.0, red = 0.1 chance level). The black contour marks the 15% accuracy threshold. A sharp boundary separates the learning phase (upper-left) from the non-learning phase (lower-right).

Key observations:

- For small batch sizes ($B \leq 512$), the network can learn effectively even when $\sim 98\%$ of pixels are corrupted. The critical corruption probability p_c remains close to 0.995.

- As batch size increases beyond ~ 1024 , the learning region shrinks sharply. At full-batch ($B = 60000$), learning fails at much lower corruption levels ($p_c \approx 0.98$).
- The transition is not gradual—accuracy drops rapidly from $\sim 70\%$ to chance level over a narrow band of p values, suggesting a genuine phase transition rather than a smooth degradation.

This reveals a striking result: **minibatch SGD noise acts as an implicit regularizer that extends learnability into regimes where full-batch training fails.**

3.2 Effect of Dataset Size

Figure 2 shows the phase diagram at each dataset size N .

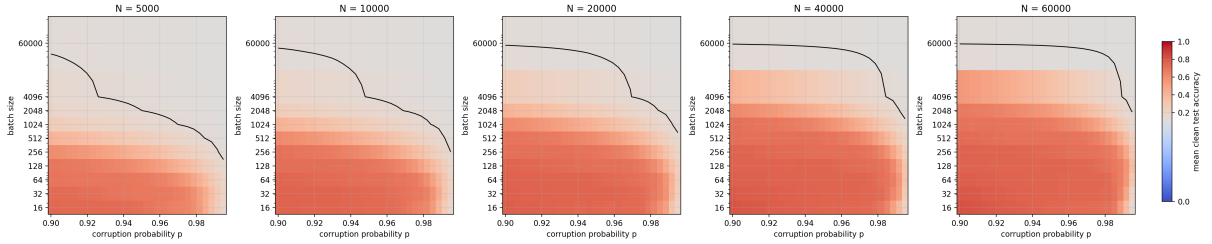


Figure 2: Phase diagrams for $N \in \{5000, 10000, 20000, 40000, 60000\}$. As N increases, the learning region expands (the phase boundary shifts rightward to higher p and downward to larger B), and the transition becomes sharper.

Key observations:

- At $N = 5000$, only small batch sizes at moderate corruption can learn. The learning region is narrow.
- With each doubling of N , the phase boundary shifts noticeably, indicating that more data helps the network tolerate greater corruption.
- The transition becomes progressively sharper with increasing N —consistent with finite-size scaling theory, where the system size (here, dataset size) controls the sharpness of the transition.

3.3 Full-Batch vs. Minibatch Comparison

Figure 3 directly compares accuracy-vs- p curves for several batch sizes at $N = 60000$.

Key observations:

- At $B = 64$, accuracy remains above 70% up to $p \approx 0.97$ and above 50% up to $p \approx 0.98$.
- Full-batch training ($B = 60000$) already begins degrading at $p \approx 0.93$ and collapses to chance by $p \approx 0.99$.
- There is a clear ordering: smaller $B \Rightarrow$ higher noise tolerance. The gap between curves widens as p increases, indicating that the regularization benefit of SGD noise grows in the high-corruption regime.

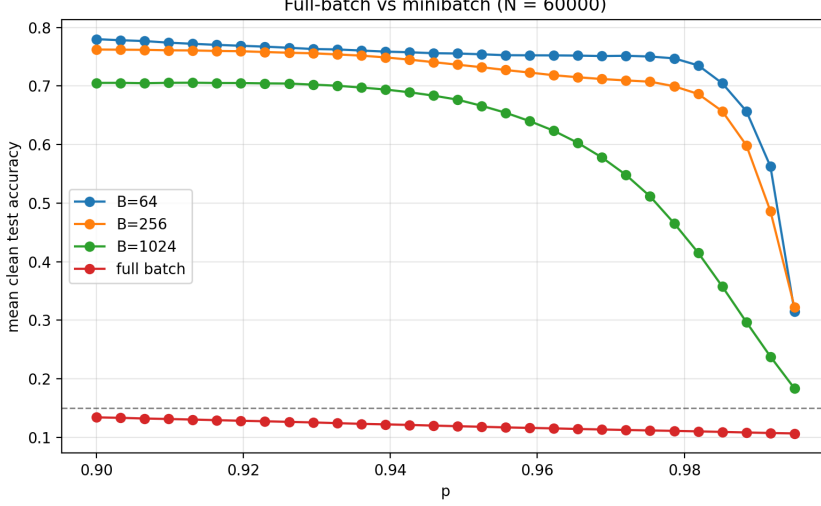


Figure 3: Mean clean-test accuracy as a function of corruption probability p for batch sizes $B \in \{64, 256, 1024, 60000\}$ at $N = 60000$. Smaller batches sustain higher accuracy deeper into the high-corruption regime.

3.4 Phase Boundary and Power-Law Fit

Figure 4 plots the estimated critical corruption $p_c(B)$ at $N = 60000$ for two accuracy thresholds.

The phase boundary is fit to the functional form

$$p_c(B) = 1 - a B^{-\beta},$$

or equivalently, the critical noise fraction scales as $(1 - p_c) \propto B^{-\beta}$. This power-law relationship suggests that the critical corruption tolerable by the network decreases algebraically with batch size.

Key observations:

- For small batch sizes ($B \lesssim 256$), p_c is nearly flat near 0.995—the network can learn even with 99.5% corruption.
- Above $B \approx 1000$, p_c drops sharply, indicating a regime change where SGD noise is no longer sufficient to regularize against input corruption.
- The power-law fit captures the overall trend, though the sharp plateau at small B suggests the true functional form may involve a saturation mechanism.

3.5 Training Dynamics Near the Phase Boundary

Figure 5 shows training loss and clean-test accuracy trajectories for two representative runs near the phase boundary.

Key observations:

- Training loss remains very close to $\ln(10) \approx 2.303$, the cross-entropy of a uniform predictor over 10 classes. This is the hallmark of near-chance performance.
- Test accuracy fluctuates between $\sim 10\%$ and $\sim 17\%$, only marginally above chance. The network extracts a faint signal but cannot robustly learn.
- The curves are noisy and non-monotone, reflecting the stochastic competition between signal and noise at the critical point.

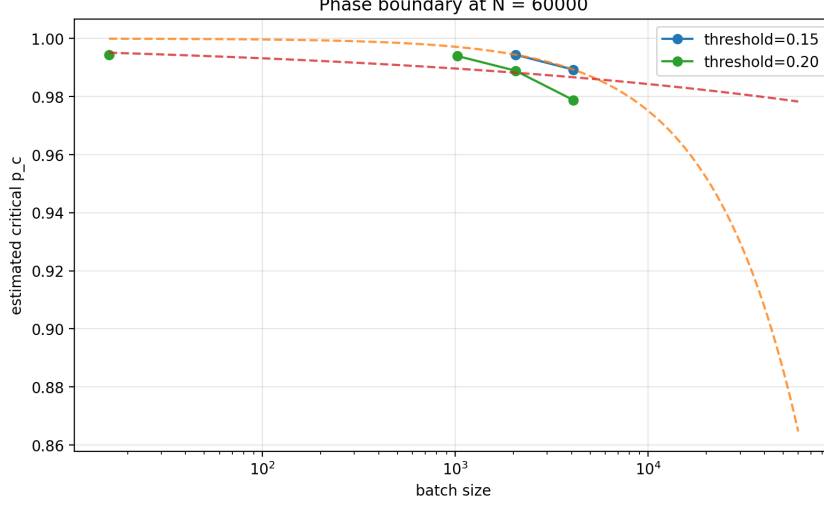


Figure 4: Estimated critical corruption probability p_c as a function of batch size at $N = 60000$, for accuracy thresholds 0.15 and 0.20. Dashed lines show power-law fits of the form $p_c(B) = 1 - a B^{-\beta}$.

3.6 Finite-Size Scaling

Figure 6 shows accuracy vs. p curves at fixed batch sizes for varying N , the analog of varying system size in a statistical-physics finite-size scaling analysis.

Key observations:

- For each batch size, the accuracy curves for different N values approximately cross at a common critical point p_c . This crossing behavior is a hallmark of a continuous phase transition.
- At $B = 64$, the crossing point is near $p_c \approx 0.985$. At $B = 1024$, it shifts to $p_c \approx 0.975$.
- The transition sharpens with increasing N : at $N = 60000$ the drop from high accuracy to chance is nearly vertical, while at $N = 5000$ it is a broad crossover.
- Larger N consistently yields higher accuracy at any fixed p , confirming that more data helps overcome corruption.

3.7 Finite-Size Scaling Collapse

Figure 7 tests whether the data from different N values collapse onto a single universal curve under the finite-size scaling ansatz:

$$A(p, N) = A_c + N^{-\kappa} \mathcal{F}((p - p_c) N^{1/\nu}),$$

where A is accuracy, p_c is the critical corruption probability, ν is the correlation-length exponent, κ governs the scaling of the order parameter at criticality, A_c is the critical accuracy, and $\mathcal{F}$ is a universal scaling function.

Key observations:

- The collapse is best at $B = 256$ and $B = 1024$, where curves from all five N values approximately overlap onto a single sigmoid-like scaling function.
- The fitted critical corruption probability p_c ranges from ~ 0.986 ($B = 64$) to ~ 0.989 ($B = 1024$), consistent with the crossing-point estimates from the raw curves.

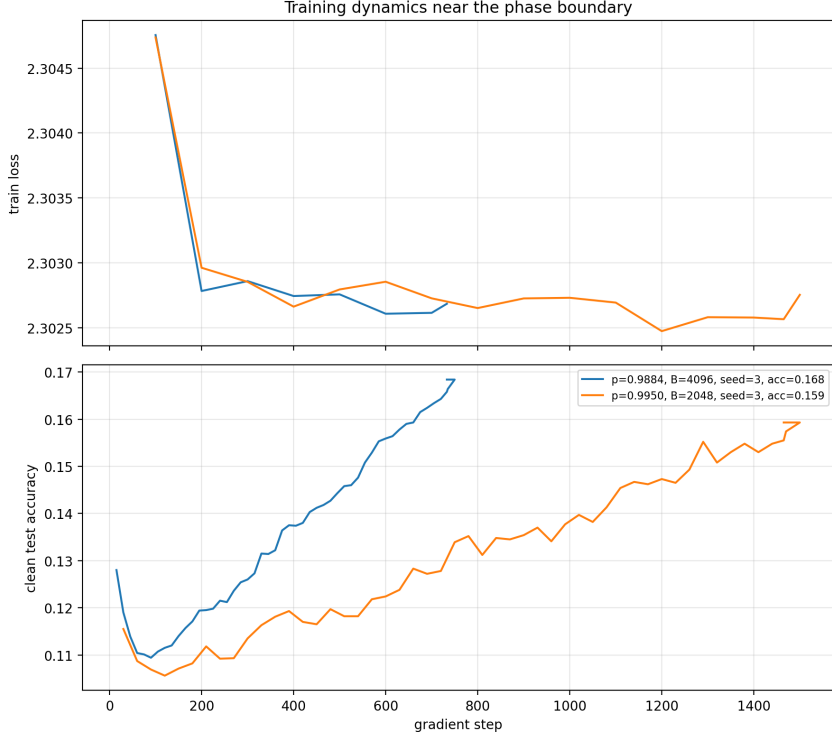


Figure 5: Training dynamics for two runs near the phase boundary ($N = 60000$). Top: training loss. Bottom: clean-test accuracy. Both runs hover near chance-level loss ($\ln 10 \approx 2.303$) and accuracy ($\sim 10\text{--}17\%$), exhibiting the marginal, noisy learning characteristic of the critical region.

- The correlation-length exponent ν ranges from ~ 0.56 to ~ 0.80 across the three batch sizes. A smaller ν indicates a sharper transition in the scaling variable.
- The exponent $\kappa = 2.0$ was found at the boundary of the search grid for all batch sizes, suggesting that the accuracy at criticality decays as N^{-2} and that the true optimal κ may lie outside the search range.
- The critical accuracy A_c ranges from ~ 0.20 to ~ 0.56 , reflecting the batch-size dependence of the order parameter at the transition.

4 Technical Description of the Experiment Script

The experiment is implemented in a single self-contained Python script (`run_experiments_2d_phase_diagram.py`) with the following structure.

4.1 Data Pipeline

1. **Base dataset:** MNIST is loaded via `torchvision` with a simple `ToTensor()` transform.
2. **Subset selection:** For $N < 60000$, a *balanced* subset is drawn: exactly $N/10$ examples per digit class, using a deterministic seed for reproducibility.
3. **Corruption:** A custom `SaltAndPepperDataset` wrapper applies per-pixel corruption on-the-fly. Each pixel is replaced with `Uniform(0, 1)` with probability p , using a seeded generator for reproducibility.
4. **Evaluation:** The clean MNIST test set (no corruption) is used for all accuracy measurements.

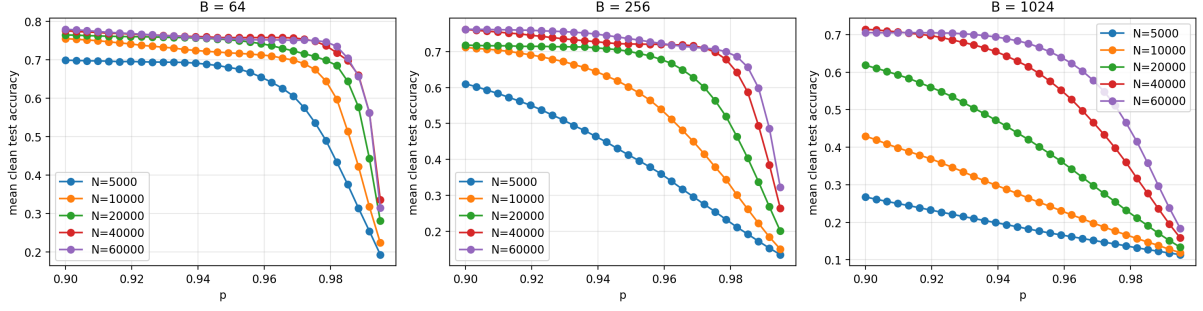


Figure 6: Finite-size scaling curves for $B \in \{64, 256, 1024\}$. Each panel shows mean clean-test accuracy vs. p for $N \in \{5000, 10000, 20000, 40000, 60000\}$. The curves steepen and converge toward a common crossing point as N increases.

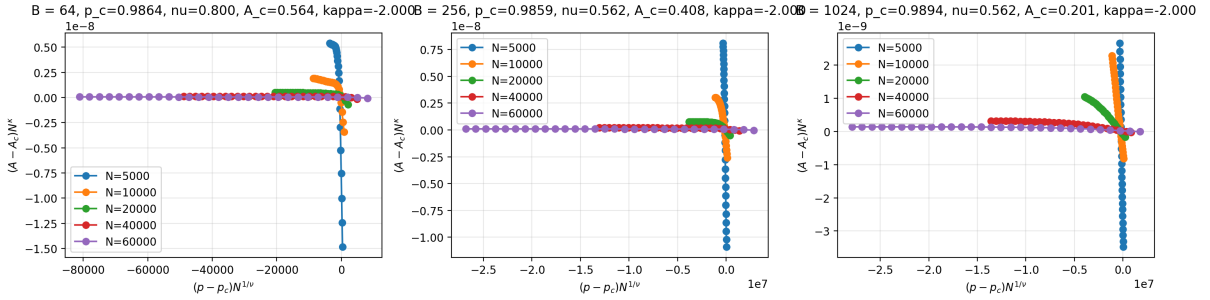


Figure 7: Finite-size scaling collapse for $B \in \{64, 256, 1024\}$. Rescaled axes: $x = (p - p_c) N^{1/\nu}$, $y = (A - A_c) N^\kappa$. The fitted critical parameters (p_c, ν, A_c, κ) are shown in each panel title.

4.2 Training Loop

The training budget is defined in terms of *total samples seen*:

$$T = N \times E_{\text{ref}} = N \times 50.$$

The number of gradient steps is then $\lceil T/B_{\text{eff}} \rceil$, where $B_{\text{eff}} = \min(B, N)$ is the effective batch size (capped at the dataset size). This ensures that changing B does not change the total amount of data the network sees—only the granularity of updates.

The optimizer is vanilla SGD with fixed learning rate $\eta = 0.01$, no momentum, no weight decay, and no learning rate schedule.

4.3 Parallelism and Execution

The script uses Python’s `ProcessPoolExecutor` to run experiments in parallel across CPU cores, with configurable worker counts and CPU affinity pinning. Each worker caches the MNIST dataset in a process-global dictionary to avoid redundant I/O. Results for each (p, B, N, seed) combination are atomically written to individual JSON files, enabling crash-safe resumption.

4.4 Aggregation and Plotting

After all runs complete, results are aggregated by computing mean and standard error of test accuracy across seeds for each (p, B, N) triplet. The script then generates the following diagnostic plots:

- **Heatmaps:** 2D color maps of accuracy in the (p, B) plane, with contour lines at the phase threshold.

- **Phase boundary:** Estimated $p_c(B)$ via linear interpolation of accuracy crossing a threshold, with power-law fits.
- **Finite-size scaling curves:** Accuracy vs. p at fixed B for varying N .
- **Scaling collapse:** Rescaled data using optimized critical exponents (p_c, ν, κ) , found via grid search minimizing the bin-variance collapse score.
- **Training dynamics:** Loss and accuracy trajectories for representative runs near the phase boundary.
- **Full-batch comparison:** Direct comparison of minibatch vs. full-batch accuracy curves.

4.5 Scaling Collapse Methodology

Two collapse methods are implemented:

Custom grid search. The script performs a brute-force search over (p_c, ν, κ) to minimize a *collapse quality score*: the data is binned along the scaled x -axis and the mean within-bin variance of the scaled y -values is computed. Lower variance $\Rightarrow$ better collapse. The search grid is:

- p_c : 121 values in a narrow window around the initial estimate.
- ν : 49 values in $[0.2, 5.0]$ (or a refined range around the initial estimate).
- κ : 81 values in $[-2.0, 2.0]$.

pyfssa library. An alternative collapse is attempted using the **pyfssa** library (finite-size scaling analysis), which implements a more principled quality-of-collapse metric. This method fixes the anomalous dimension $\zeta = 0$ and searches over (p_c, ν) only. In the current experiment run, this method failed due to a NumPy compatibility issue.

5 Summary

This experiment establishes that the transition from learning to non-learning under heavy pixel corruption in MNIST exhibits the hallmarks of a *continuous phase transition*:

- A sharp boundary in the (p, B) control-parameter space.
- Finite-size scaling: the transition sharpens as dataset size N increases.
- Approximate data collapse under a scaling ansatz with well-defined critical exponents.
- A common crossing point in accuracy curves at different N , identifying the critical corruption p_c .

The most striking practical finding is the role of batch size: **smaller batches dramatically extend the learnable regime**, allowing networks to extract signal even when 99%+ of pixels are noise. This suggests that the implicit regularization provided by minibatch SGD stochasticity is not merely a convenience but a qualitatively important factor that shifts the phase boundary of learnability.